Making Scientific Knowledge Navigable for Agents

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MLOps Community · Research Copilot Workshop

Researchers drown in literature. Agents could help — ~~but~~ not with today's tools.

3M+

scientific papers published per year

and doubling every ~12 years

20–30%

of a researcher's time

spent on literature search

~10

results per search query

no graph, no memory, no structure

A research copilot built on vanilla search is stuck with the same primitives as the human — scaled up.

What "navigable" means for an agent

Agents need different primitives than human searchers

Human searcher

Ranked list

Scans titles, opens a few, satisfied.

One-shot query

Session state lives in the researcher's head.

Visual cues

Journal, authors, year, formatting.

Stops when tired

Implicit budget, intuition-driven.

Agent

Graph topology

Which papers bridge communities? Which are foundational?

Session working set

Accumulates, dedupes, tags, reasons across turns.

Structured returns

Bibcodes, IDs, community labels, provenance.

Multi-hop moves

Citation chains, co-citation, find_gaps, temporal.

What I'll cover

30 minutes · 5 sections · live demo

1 **The problem space**
Search APIs vs. agent needs. Why MCP.

2 **Corpus completeness**
The 17.8% → 99.6% result that changes everything.

3 **Architecture**
PostgreSQL + pgvector + 13 MCP tools on a single box.

4 **Live demo**
Agent maps a research landscape with hybrid retrieval + graph moves.

5 **Hard-won lessons**
Embedding pipeline, UNLOGGED disaster, surprising eval results.

Search APIs return ranked lists. Agents need a knowledge structure.

What's missing when you wrap an LLM around /search

No graph topology

No way to ask "which papers bridge these two communities?" or "which papers are structurally central?"

No session memory

Every query starts cold. Working sets live in the prompt or nowhere.

No community structure

Can't browse the research landscape, can't see the lens you're navigating through (citation vs. semantic vs. taxonomic).

No multi-model signal

One embedding, one ranker. Dense-only fails on specific-term queries; lexical-only fails on paraphrase.

No structured returns

Free-text snippets burn context and break composition with other tools.

Why MCP

Model Context Protocol is a good fit for agent-native knowledge

Tool-calling primitives

First-class: name, schema, typed args, typed returns. The agent knows what it can do.

Session context

Stateful per-connection. Working sets live on the server, not in the prompt.

Structured responses

JSON objects, not HTML snippets. Compose cleanly with other tools.

Transport agnostic

stdio, HTTP, WebSocket. Works with Claude Desktop, Cursor, custom agents.

Portable discipline

If your research copilot talks MCP, the same knowledge layer can be hosted by ADS, arXiv, or you.

The corpus

NASA ADS — the canonical bibliographic database for astronomy

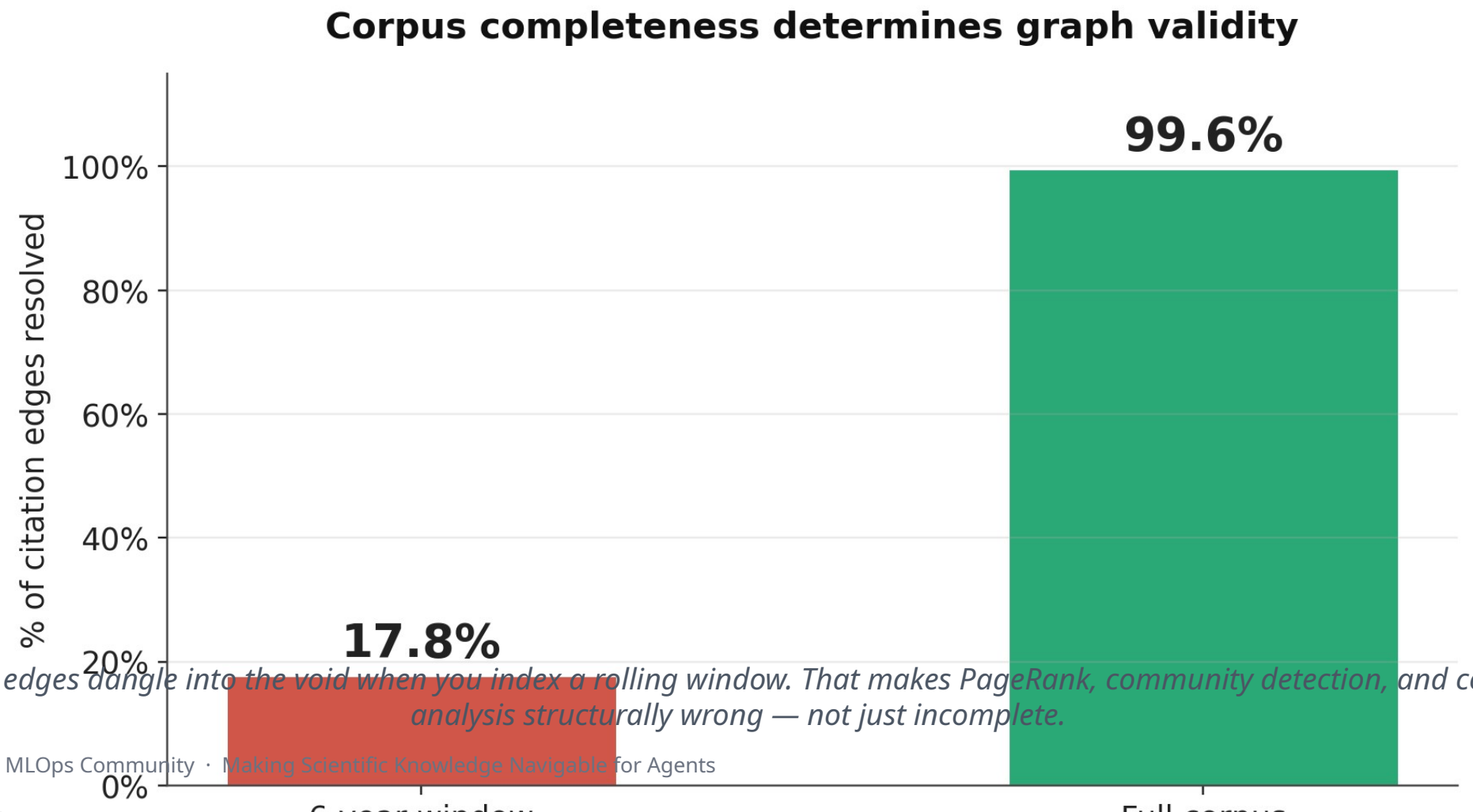
Corpus at a glance

32.4M	299M	32.4M
Papers	Citation edges	INDUS embeddings
1800–2026	*99.6% resolved*	*full coverage*

14.9M papers with full body text (46%) · PostgreSQL 16 + pgvector · single server (62 GB RAM)

Corpus completeness determines graph validity

A 6-year window resolves 17.8% of citation edges. The full corpus resolves 99.6%.



82.2% of edges dangle into the void when you index a rolling window. That makes PageRank, community detection, and co-citation analysis structurally wrong — not just incomplete.

From 45K fragmented components to a single giant component

6-year window

2021–2026, ~5M papers

~45,000	disconnected components
~1.7M	isolated nodes
17.8%	of edges resolved
misleading	PageRank, Leiden, co-citation

Full corpus

1800–2026, 32.4M papers

19.98M	in the giant component (99.3% of connected)
36	size of the 2nd-largest component
99.6%	of edges resolved
valid	PageRank, Leiden, co-citation

Only 1 connected component exceeds 100 nodes — the giant component itself.

Takeaway: corpus completeness is a retrieval-quality axis

Not a "nice to have" — a prerequisite for agentic graph reasoning

Rolling-window bias

Recent papers cite older papers. A rolling window systematically excludes the very papers its contents reference.

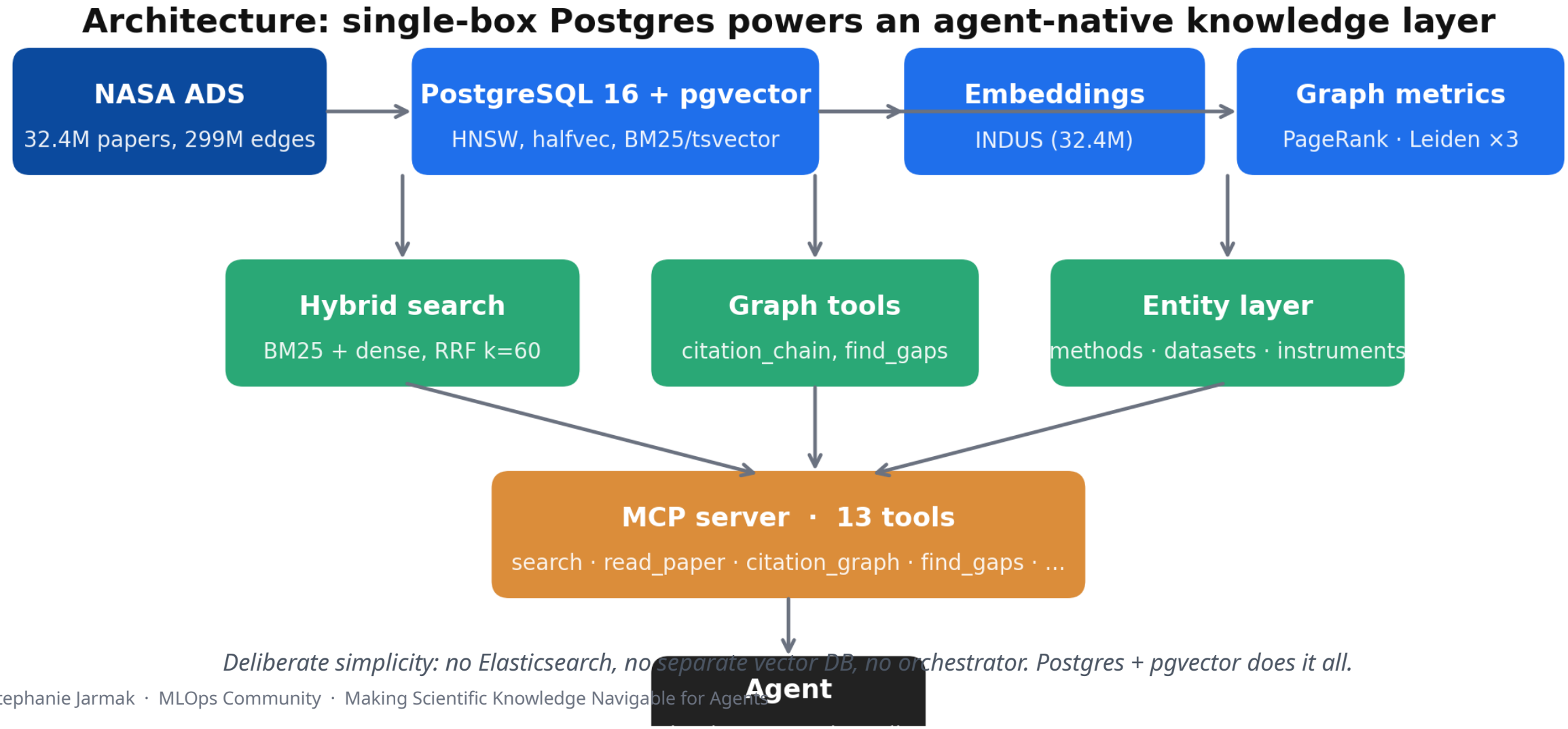
Full-corpus ingest

Feasible. 32.4M papers into PostgreSQL via COPY in ~4 hours on a single machine. No excuse.

Copilot design

Treat your corpus boundary as a product decision, not an operational one. It shapes what the agent can reason about.

Architecture: one PostgreSQL box, three layers, one MCP server



Deliberate simplicity: no Elasticsearch, no separate vector DB, no orchestrator. Postgres + pgvector does it all.

Retrieval stack: three signals, reciprocal rank fusion

Dense (domain) + dense (query-doc) + lexical (BM25)

INDUS

nasa-impact/nasa-smd-ibm-st-v2

768d · CLS pool on title+abstract

32.4M papers embedded

Domain-tuned on 2.66M ADS pairs

Local, open-weight, GPU only

text-embedding-3-large

OpenAI, 3072d → 1024d Matryoshka

Query-to-document retrieval

Complements INDUS on NL queries

Circuit-breaker fallback

Halfvec quantization

BM25

tsvector on title+abstract+keywords

Native PostgreSQL GIN index

Catches specific-term queries

Cheap, interpretable

Zero infra cost

Reciprocal Rank Fusion (k=60) → hybrid_search

pgvector + Qdrant: complementary, not competing

Same corpus, different jobs — when each wins

pgvector — source of truth

why it won the main path

- Same transaction as citation_edges, paper_metrics
- 32M rows, 299M edges — one query plan
- No sync layer, no eventual consistency
- Halfvec + HNSW fits in one Postgres box
- Ops team already runs it

Qdrant — enrichment shard

where it adds real value

- Named multi-vectors per point (INDUS + query-doc)
- Payload-indexed filters at HNSW speed
- PQ / SQ / binary quantization + rerank
- Recommendation / discovery API as a primitive
- GPU HNSW build, per-collection snapshots

Pragmatic shape: Postgres = source of truth · Qdrant = additive search/discovery layer, fed by one-way upsert

Prototype: find_similar_by_examples (14th MCP tool)

"More like these papers, less like those" — optionally filtered

```
find_similar_by_examples(
    positive_bibcodes: list[str],
    negative_bibcodes: list[str] = [],
    limit: int = 10,
    year_min / year_max: int | None,
    doctype: list[str] | None,
    community_semantic: int | None,
    arxiv_class: list[str] | None,
)
```

Pilot state

400,000 points	*top-PageRank, rich payload*
INDUS 768d	*cosine similarity, HNSW*
~676 pts/s	*upsert from Postgres*
2.6 GB on disk	*one Docker container*

Live demo — seed: two graphene/terahertz papers

1. **2012NJPh...14k8001T c=11 physics.optics s=0.721**
   Comment on 'Encoding many channels on the same frequency through radio vorticity...'
2. **2010ACSNa...4.1889F c=5 cond-mat.mes-hall s=0.676**
   Thermal Conductivity of Graphene in Corbino Membrane Geometry
3. **2012NatMa..11..865V c=15 cond-mat.mes-hall s=0.659**
   Graphene FETs as room-temperature terahertz detectors
4. **2015NatNa..10..437T c=15 cond-mat.mes-hall s=0.658**
   Photovoltage in graphene on a femtosecond timescale

community_semantic=15 filter → restrict to graphene/cond-mat cluster. Negative example → push results away from that direction. All three modes, one tool call.

Graph intelligence: topology, communities, multi-hop moves

Computed once on the full graph — served as columns on paper records

Per-paper metrics

- PageRank on 32.4M nodes
- HITS hub / authority
- Three Leiden resolutions
- Computed weekly, ~10 min
- Stored as columns, not offline

Three community lenses

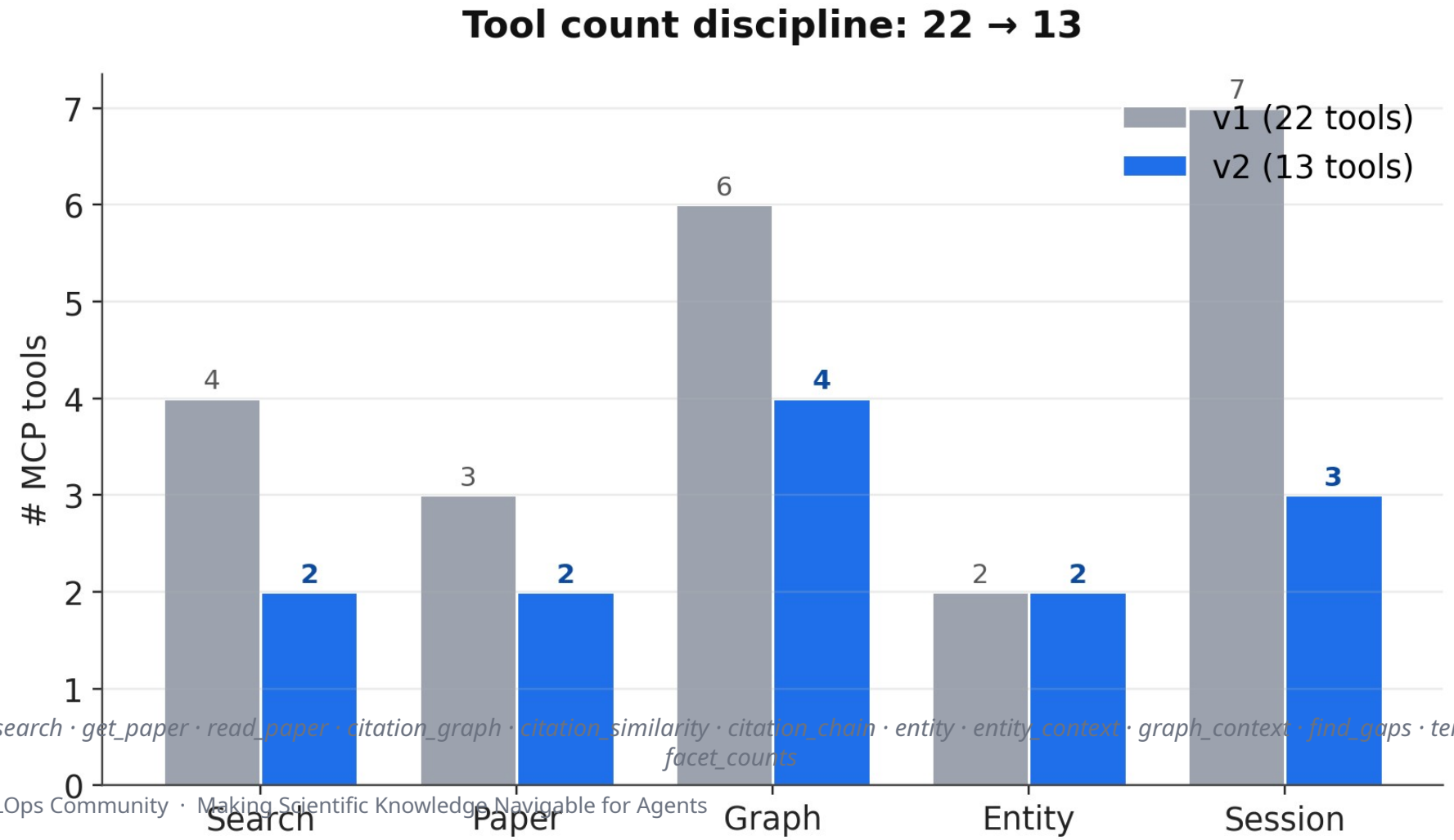
- Citation (Leiden) — who cites whom
- Semantic (k-means on INDUS) — similar content
- Taxonomic (arXiv / UAT) — same field
- Disagreements are informative
- Agent picks the lens

Multi-hop tools

- citation_chain (shortest path)
- citation_similarity (co-cite / biblio-couple)
- temporal_evolution
- find_gaps (bridge papers)
- graph_context (neighborhood)

13 MCP tools, down from 22

Premortem: >15 tools degrades agent selection accuracy



search · concept_search · get_paper · read_paper · citation_graph · citation_similarity · citation_chain · entity · entity_context · graph_context · find_gaps · temporal_evolution · facet_counts

Tool design for agents ≠ tool design for humans

Five principles from watching Claude burn context unnecessarily

Return compact stubs	bibcode · title · first_author · year · citation_count · snippet. Full paper via read_paper only when needed.
Stable, typed interfaces	JSON schema. No free-text fields. Every response parses.
Include provenance	Which tool returned this? Which query? Agent can reason about its own trail.
Idempotent reads	Same inputs → same outputs. Caching is trivial.
Cursor-based pagination	Big result sets return a cursor, not 10,000 bibcodes.

Session state: the agent's short-term memory

In-memory working set per MCP connection — not in the prompt, not in a DB

```
WorkingSetEntry(
    bibcode="2019A&A...622A...2P",
    source_tool="hybrid_search",
    source_context={
        "query": "exoplanet atmospheres",
        "rank": 3,
        "community": "coarse_14"
    },
    tags=["foundational", "review"],
    added_at="2026-04-22T14:03:19Z",
)
```

What it unlocks

find_gaps

Bridge papers between working-set communities not yet in the set

get_session_summary

What the agent knows so far, grouped by community

Dedup and provenance

Don't re-add. Remember how we found this.

Cross-tool reasoning

Combine a hybrid_search result with a citation_chain hit

Three community lenses — and their informative disagreements

Each lens has clean semantics. The agent picks.

Citation

behavioral affinity

- Leiden on giant component
- 3 resolutions: ~20 · ~200 · ~2000
- Coverage: 62% of papers
- "Papers that cite each other"
- Finds invisible-college patterns

Semantic

topical affinity

- k-means on INDUS 768d
- 3 resolutions
- Coverage: 99% (all INDUS-embedded)
- "Papers with similar content"
- Catches cross-citing subfields

Taxonomic

institutional affinity

- arXiv class + UAT concepts
- Hierarchical, curated
- Coverage: ~8% (arXiv subset)
- "Papers in the same field"
- Sanity-check for the other two

Live demo

A research copilot maps the exoplanet-atmospheres landscape

User prompt

"I'm preparing a grant proposal on exoplanet atmospheric characterization. Map the research landscape — what are the major subcommunities, which papers are foundational, where are the recent bridges between communities, and what am I missing?"

What the copilot needs

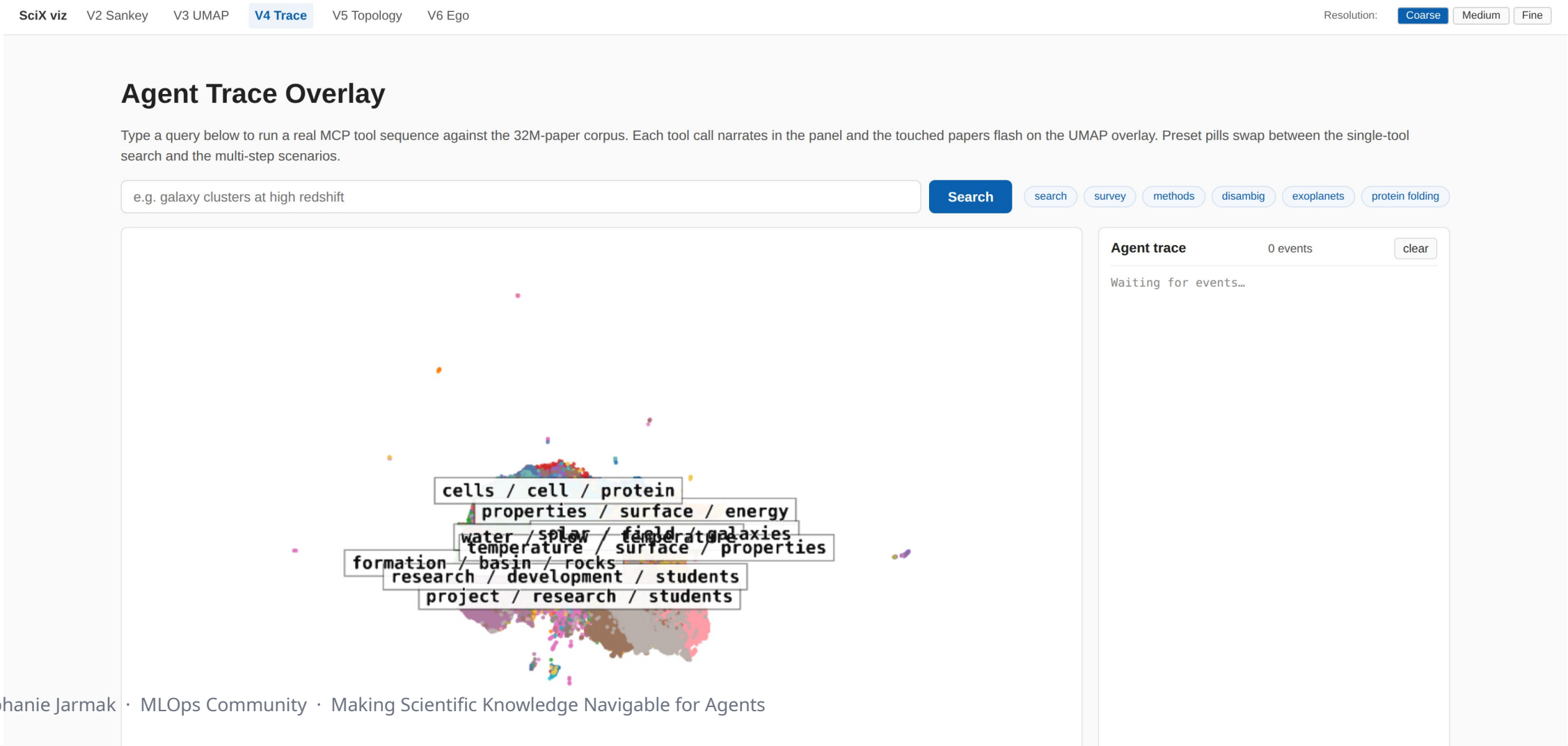
Topical search · community structure · PageRank within community · temporal signal · bridge-paper detection · a memory to reason over.

Tools invoked (order)

1	**search**	broad hybrid hit on "exoplanet atmospheric characterization"
2	**graph_context**	pull community assignments + PageRank on top hits
3	**citation_similarity**	expand with co-citation bridges
4	**temporal_evolution**	see which communities are growing
5	**find_gaps**	bridge papers between working-set communities
6	**read_paper**	deep-dive on 3 foundational papers

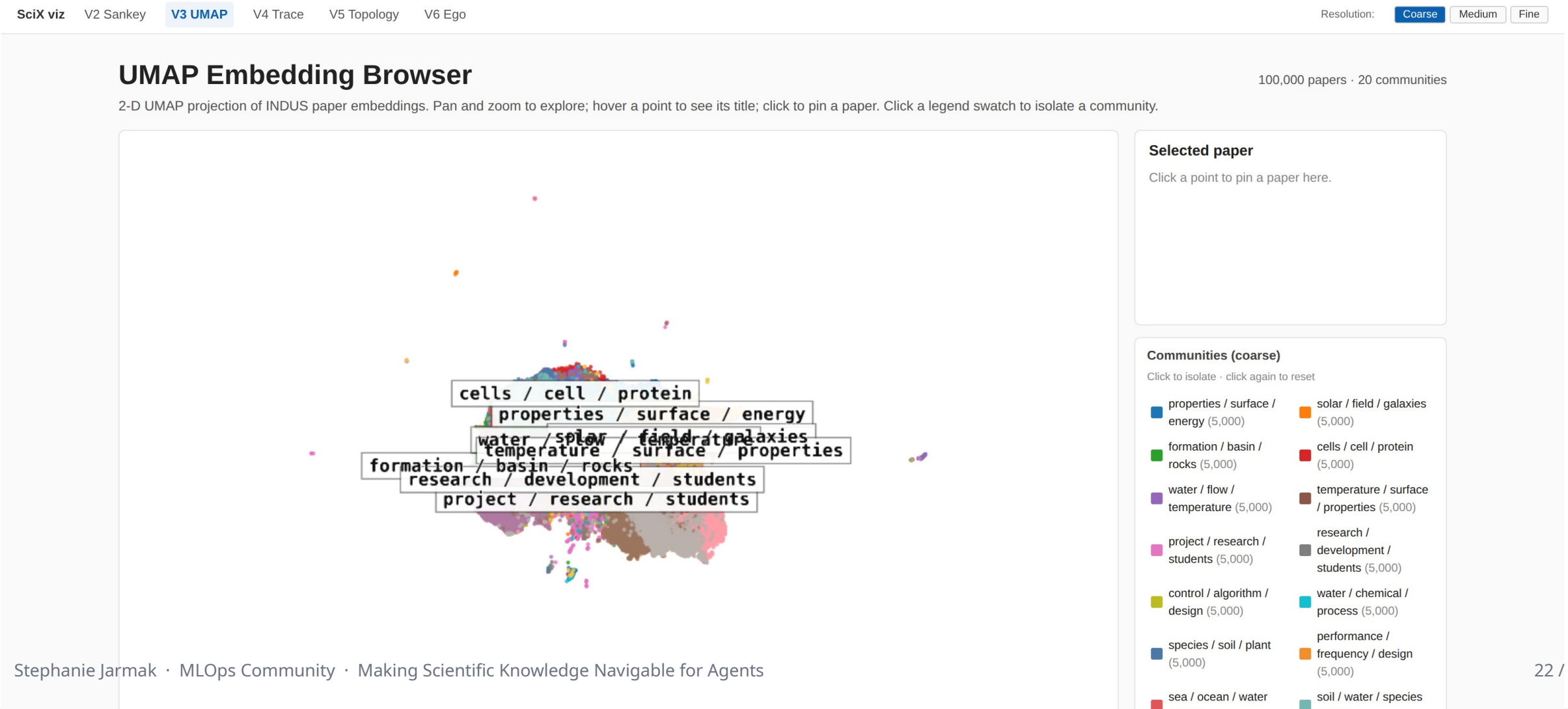
Agent trace: tool calls visualized over time

web/viz/agent_trace — live narration of the MCP session



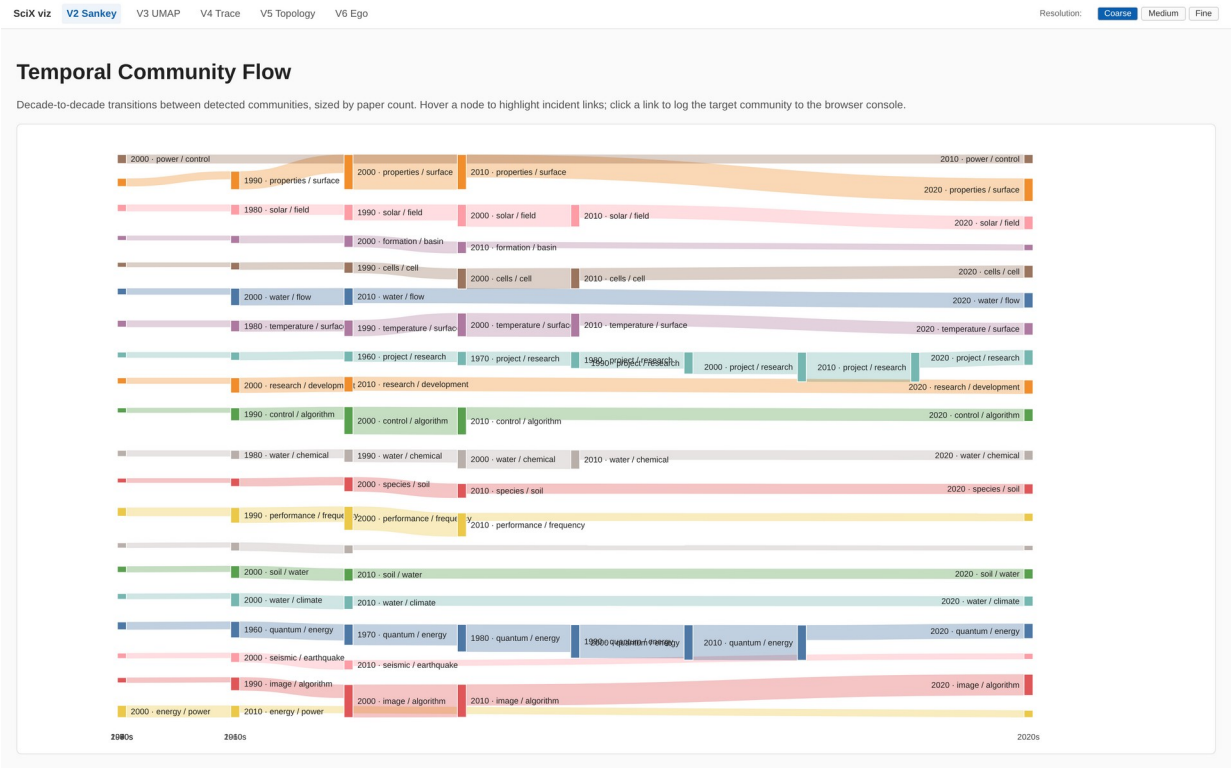
UMAP of 32M INDUS embeddings — the research landscape the agent navigates

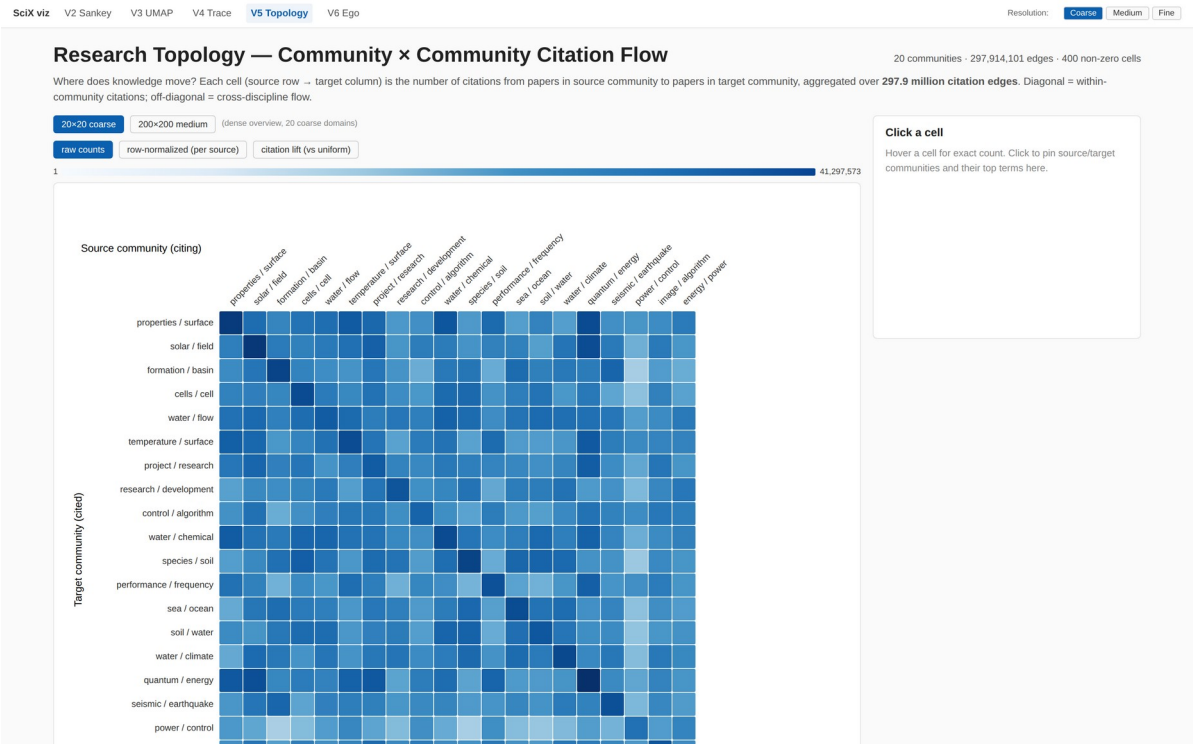
Communities color-coded · brightness = PageRank · hover for bibcode



Citation flow between communities

Sankey: how citations move across Leiden communities · Heatmap: 199×199 community co-citation

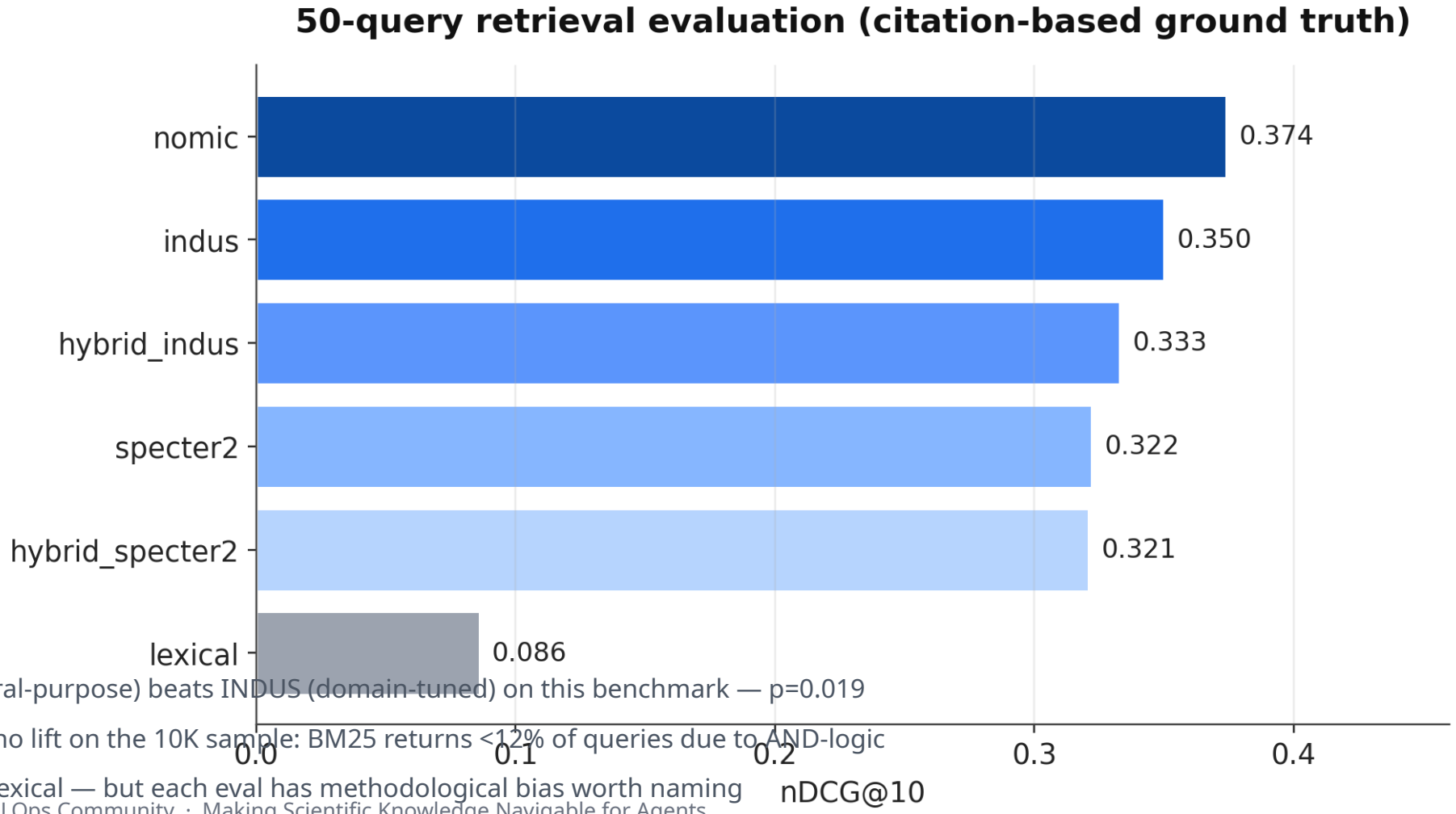




These are the structural views an agent reasons over — not visually, but as graph edges and community IDs.

Retrieval evaluation — and a surprise

50 queries · citation-based ground truth · latest rebaseline



- Nomic (general-purpose) beats INDUS (domain-tuned) on this benchmark — p=0.019
- Hybrid adds no lift on the 10K sample: BM25 returns <12% of queries due to AND-logic
- All dense ≫ lexical — but each eval has methodological bias worth naming

Surprise: general-purpose nomic beats domain-tuned INDUS. Evaluation methodology matters.

Lesson 1: never use UNLOGGED tables for data you can't reconstruct

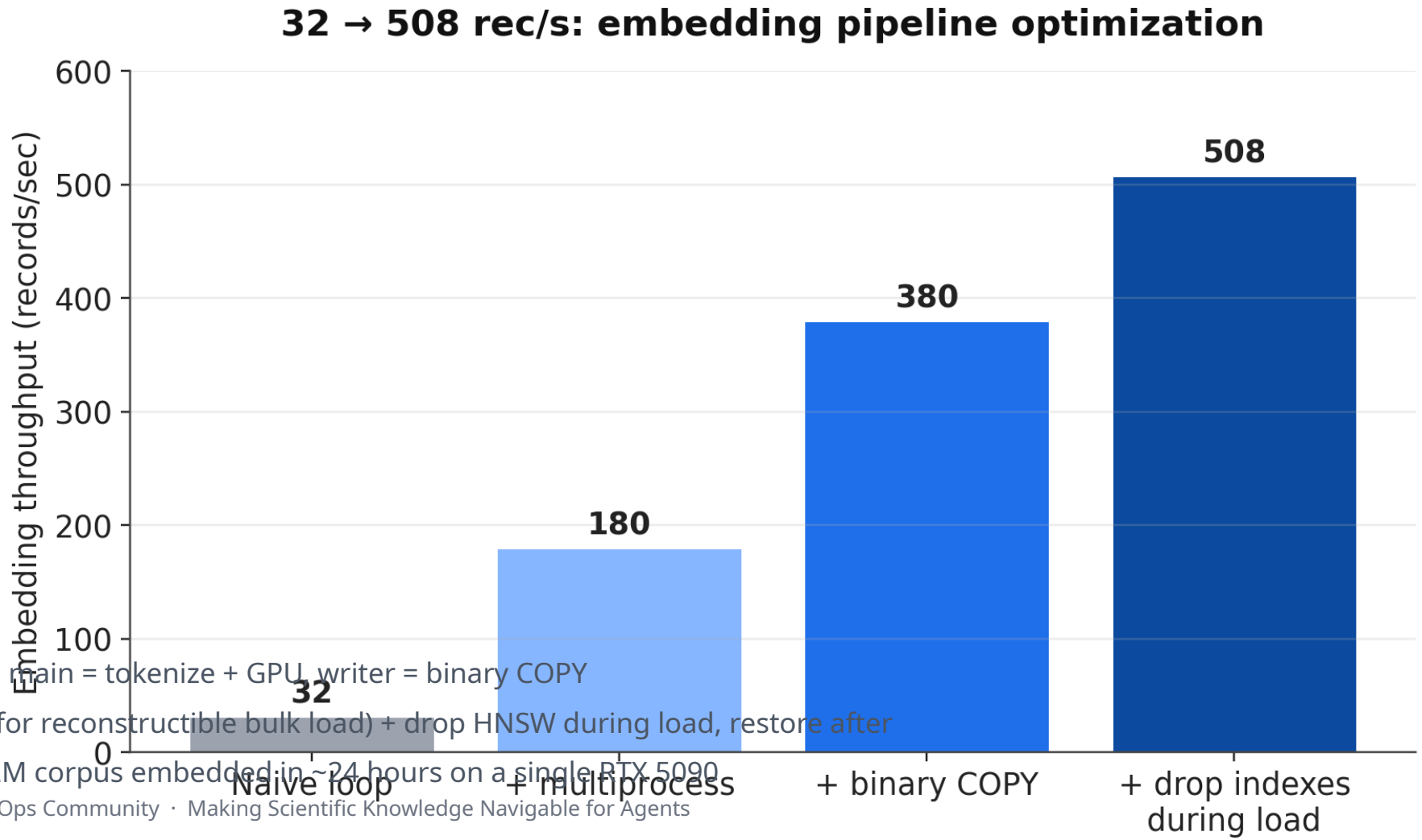
We lost 32M SPECTER2 embeddings to a Postgres restart

Day 1–3	Bulk-embed 32M papers using UNLOGGED paper_embeddings (~15% faster writes)
Day 4	Postgres OOMs due to an unrelated query
Day 4 + 1min	systemd restarts Postgres — UNLOGGED tables are truncated at startup
Day 4 + 2min	32 million embeddings, ~100 GPU-hours, gone
Day 5+	Rebuild. Migration 023 changes paper_embeddings to LOGGED permanently.

If your pipeline builds something expensive, it must survive a process restart. UNLOGGED is a write-optimization for reconstructible data only — never persistent state.

Lesson 2: the gap between naive and tuned pipelines is 10–20×

Embedding throughput, same hardware



- Multiprocess: main = tokenize + GPU, writer = binary COPY
- UNLOGGED (for reconstructible bulk load) + drop HNSW during load, restore after
- Result: full 32M corpus embedded in ~24 hours on a single RTX 5090

Lesson 3: LLM cost estimates are 3–5× too low

Prompt overhead dominates output tokens for structured extraction

Entity extraction on 10K papers (Claude Haiku)

Full 32M corpus projection

Why: a 500-token prompt extracting 50 tokens of structured output has 10× overhead. Caching recovers some; fine-tuning a smaller model may be necessary at full scale.

What's next

Research copilot gets deeper as the knowledge layer gets richer

Chunk-and-embed full text	14.9M bodies already ingested · contextual retrieval (Anthropic pattern) · 35% reduction in retrieval failure projected
Graph-RAG	HippoRAG 2-style personalized PageRank over extracted concepts · 7% lift over dense embeddings at 10-30x lower cost
Better evaluation	200+ query benchmark · expert-judged graded relevance · agent-task evaluation, not just retrieval
Upstream to ADS	Embeddings and graph metrics as data products · MCP endpoint alongside REST API · no re-harvest needed
Multi-corpus	Same MCP surface over arXiv, bioRxiv, APS · agent writes corpus-agnostic code

A checklist for research copilot builders

- ✓ **Corpus boundary** — Set it deliberately. Partial-corpus graph analytics are structurally wrong, not just incomplete.
- ✓ **Hybrid retrieval** — Dense (domain) + dense (query-doc) + lexical. RRF fuses. Each alone has predictable failure modes.
- ✓ **Graph primitives** — PageRank, communities, citation chains — as first-class MCP tools, not offline batch.
- ✓ **Session memory** — Working set lives on the server. Provenance tags. Not in the prompt.
- ✓ **Tool count discipline** — < 15 tools. Compact stubs. Stable schemas. Cursor pagination.
- ✓ **Own your eval** — 50 queries minimum. Name your methodology bias. Validate with experts.
- ✓ **Instrument everything** — Agent-trace viz is the fastest way to learn whether your tools are working.

Thank you

Questions?

Stephanie Jarmak

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Resources

- Repository — github.com/sjarmak/scix-agent
- Live MCP endpoint — mcp.sjarmak.ai/mcp/
- Paper (ADASS 2026, preprint) — arXiv (pending)
- Contact — steph.jarmak@gmail.com